

## Course Outline

### 1. Course Identity

#### A. Course as listed in CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

|                         |                                                                                                                                                                                                                                                                                          |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Course code             | DDA5002                                                                                                                                                                                                                                                                                  |
| Course title (English)  | Optimization                                                                                                                                                                                                                                                                             |
| Course title (Chinese)  | 最优化                                                                                                                                                                                                                                                                                      |
| Units                   | 3                                                                                                                                                                                                                                                                                        |
| Language of Instruction | English                                                                                                                                                                                                                                                                                  |
| Description (English)   | This course teaches basic theory and methodologies for optimization and demonstrates applications in data science, financial engineering, machine learning, etc. The course covers models and solution algorithms of linear programming, nonlinear programming, and integer programming. |
| Description (Chinese)   | 本课程教授最优化的基本理论和方法，并演示其在数据科学、金融工程以及机器学习等领域中的应用。本课程包括线性规划、非线性规划和整数规划中的建模和求解算法。                                                                                                                                                                                                              |

#### B. Corresponding course in CUHK

Please give details of the *closest* corresponding course in CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course in CUHK, please make multiple copies of the block below.

|                        |                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Course code            | MMAT5240                                                                                                                                                                                                                                                                                                                                                                                           |
| Course title (English) | Optimization & Modeling                                                                                                                                                                                                                                                                                                                                                                            |
| Course title (Chinese) | 优化与建模                                                                                                                                                                                                                                                                                                                                                                                              |
| Units                  | 3                                                                                                                                                                                                                                                                                                                                                                                                  |
| Description (English)  | This course is designed for the M.Sc. Degree Programme in Mathematics. The course provides an introduction to optimization methods, optimization theory and modeling. Topics will be selected from constrained and unconstrained optimization, descent methods, Newton type methods, calculus of variations, linear and non-linear programming, applications to "real life" problems and modeling. |
| Description (Chinese)  | 本课程为数学硕士学位课程而设。本课程主要介绍优化方法、优化理论和建模。主要将从有约束和无约束优化、下降法、牛顿型方法、变分微积分、线性和非线性规划、应用到"现实生活"问题和建模中选择。                                                                                                                                                                                                                                                                                                       |

## 2. **Prerequisites / Co-requisites**

Please state prerequisites and co-requisites, in terms of courses in CUHK-Shenzhen\* or any other requirements (e.g., having taken certain subjects in high school).

(\* Because course codes may not yet be stable, please provide both course code and course title.)

### A. **Prerequisites**

MAT1001 Calculus I    MAT1002 Calculus II    MAT2040 Linear Algebra

### B. **Co-requisites**

## 3. **Learning Outcomes**

The objective of this course is for students to learn the most important types of mathematical optimization models, and how to formulate and solve them. Specifically, students will:

- know and be able to formulate optimization models for real-world problems;
- be able to explain the properties of optimization models;
- duality and sensitivity analysis and be able to use those concepts to predict What-If scenarios;
- know and be able to apply several fundamental optimization algorithms;
- be able to model and solve optimization problems with mathematical optimization software;
- build problem-solving skills through critical thinking and the application of optimization techniques to practical scenarios.

## 4. **Course syllabus**

The course topics include formulation of optimization models, algorithms for solving these models, and requisite theory, with specific applications in data science, financial engineering, and machine learning. Specific types of models include linear programming, nonlinear programming, and integer programming. Specific solution algorithms include: the simplex method for linear programs; specialized network optimization algorithms; the branch-and-bound method for integer programs; and improving-direction methods for

nonlinear programs. Some topics in theory arise naturally, as needed, when developing optimization algorithms. Further topics in theory include sensitivity analysis and duality for linear programs. An open-source optimization modeling language based on Python, along with associated solvers, will be used to implement and numerically solve optimization models.

## 5. Assessment Scheme

| Component/ method | % weight |
|-------------------|----------|
| Homework          | 30       |
| Midterm Exam      | 35       |
| Final Exam        | 35       |

## 6. Grade descriptor

### General

| Grade | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A     | Demonstrates a thorough knowledge and understanding of the principles of optimization; successfully applies optimization methods at a sophisticated level in a wide variety of contexts; successfully transform a verbal description of a given decision problem to a meaningful mathematical model; makes generalizations and justifies conclusions; understands and interprets the solution obtained from solving an optimization problem; communicates optimization methods in a clear, effective and concise manner, using correct techniques, notation and terminology; demonstrates the ability to integrate knowledge, understanding and skills from different areas of the course; uses technology proficiently. |
| A-    | Demonstrates a broad knowledge and understanding of the principles of optimization systems; successfully applies optimization methods in a variety of contexts; successfully transform a verbal description of a given decision problem to a meaningful mathematical model; makes generalizations and justifies conclusions; understands and interprets the solution obtained from solving an optimization problem; communicates optimization methods in a clear and effective manner, using correct techniques, notation and terminology; demonstrates some ability to integrate knowledge, understanding and skills from different areas of the course; uses technology proficiently.                                  |
| B     | Demonstrates a good knowledge and understanding of the principles of optimization; successfully applies optimization methods in performing routine task; successfully transform a verbal description of a given decision problem to a meaningful mathematical model in routine situations; understands the solution obtained from solving an optimization problem; communicates optimization methods using suitable notation and terminology; demonstrates an awareness of the links between different areas of the course; uses technology appropriately.                                                                                                                                                               |
| C     | Demonstrates a satisfactory knowledge and understanding of the principles of optimization; successfully transform a verbal description for some routine problems to a mathematical model; successfully carry out mathematical techniques to analyze and                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|   |                                                                                                                                                                                                                                                                                                                   |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | interpret the performance of a communication system in straightforward contexts; has limited ability to understand the solution obtained from solving an optimization problem; communicates optimization methods adequately; uses some appropriate notation and terminology; uses technology satisfactorily.      |
| D | Demonstrates partial knowledge and limited understanding of the principles of optimization; attempts to transform a verbal description for some routine problems to a mathematical model; communicates some optimization methods; uses appropriate notation and terminology; uses technology to a limited extent. |
| F | Demonstrates minimal knowledge of the principles of optimization; demonstrate little or no ability to apply optimization methods in routine tasks; is unable to make effective use of technology.                                                                                                                 |

## 7. Feedback for evaluation

Formal Course and Teaching Evaluation

Informal feedback to instructor and/or teaching assistant

Departmental retreat and programme review

## 8. Reading

### Recommended Reading:

1.Introduction to Linear Optimization, Bertsimas, D., Tsitsiklis, J., Athena Scientific, 1997

2.Convex Optimization, Boyd, S., Vandenberghe, L, Cambridge University Press, 2009

3.Introduction to Nonlinear Optimization, Theory, Algorithms, and Applications with MATLAB, Beck, A., SIAM-Society for Industrial and Applied Mathematics, 2014

## 9. Course components

| Activity | Hours/week |
|----------|------------|
| Lectures | 3          |

|           |   |
|-----------|---|
| Tutorials | 1 |
|-----------|---|

### 10. Indicative teaching plan

| Week | Content/ topic/ activity |
|------|--------------------------|
|------|--------------------------|

### 11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections.

[Example: 150 students for lecture (x 2); 30 students for tutorials (x 10)]

Approximately 120 students for three sections of lecture; 60 students for six sections of tutorials. (360 total)

### 12. Any other information

### 13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences?

Have the details (as in this document) been approved at School or other level in CUHK-Shenzhen?

Submit application

### 14. Version date

|                |            |
|----------------|------------|
| Version number | 1          |
| As of (date)   | 2025-07-25 |